

Task 2 (100 points)

1. Problem Statement

- Implement an OpenMP-parallelized version of the function $f(x) = 4/(1+x^2)$.
- Use `integrate_omp_atomic` to compute the integral. Use `#pragma omp atomic` to ensure thread safety.
- Generate a table of results in `tables/integration_summary.csv`, and a speedup graph in `graphs/task2_speedup.pdf`.

2. Implementation Details

- The number of steps is fixed to `nsteps = 40000000`.
- The number of processors is fixed to 1, 2, 4, 7, 8, 16, 20, 40.
- The results should be stored in a table with columns for the number of processors, the execution time, and the speedup.

3. Submission

Submit your code and the generated files to the submission portal. The deadline is 2024-05-15 23:59:59. Late submissions will not be accepted.